

Replication Report:
Ferroelectricity, SSFLC, bistability and all that
(I. Dahl, arXiv:cond-mat/0211693, 2002)

Automated replication — texture class: **polar** (SSFLC / smectic-C*)

2026-07-19

1 Nature of the source paper

The target is **not a primary research paper**. It is a critical book-review / terminology-and-priority polemic of S. T. Lagerwall’s book *Ferroelectric and Antiferroelectric Liquid Crystals* (1999). The bulk of the text disputes the definitions of “polar,” “ferroelectric,” “SSFLC,” “hysteresis” and “bistability,” contests historical priority for the discovery of ferroelectric liquid crystals (Meyer 1974 vs. Clark–Lagerwall), and criticises the pictorial evidence in the book. It contains **no self-contained solvable model, no dataset, and no reproducible figure**. The scaffold’s method extract correctly flagged it NON-REPLICABLE AS POSED.

Rather than fabricate a result, we **reduced the physics the paper argues *about*** — the standard Clark–Lagerwall surface-stabilized ferroelectric liquid-crystal (SSFLC) picture, plus Dahl’s own qualitative “alternative view” — to a minimal, tractable azimuthal (ϕ) director model on the smectic cone, and made five of its statements machine-checkable. This is a *physics-of-the-domain* replication, not a reproduction of a numerical result the paper never reports.

2 Minimal model

In the chiral smectic-C* phase the director $\mathbf{n}$ lies on a cone of fixed tilt (cone half-angle) θ about the layer normal; its state is the azimuth $\phi(z)$ across the cell. The spontaneous polarization $\mathbf{P} \perp$ tilt plane, $|\mathbf{P}| = P_s$, couples to a field E along the cell normal. Free energy per unit area:

$$F = \int_0^d \left[\frac{K}{2} (\partial_z \phi)^2 - P_s E \sin \phi - \frac{1}{2} \varepsilon_0 \Delta \varepsilon E^2 \sin^2 \theta \cos^2 \phi \right] dz + W_s [g(\phi_0) + g(\phi_d)]. \quad (1)$$

Representative SSFLC material parameters (DOBAMBC/HOBACPC-like, from the literature the paper cites): $K = 5 \text{ pN}$, $P_s = 40 \text{ nC/cm}^2$, $\theta = 22.5^\circ$, $\gamma = 0.1 \text{ Pa s}$, $d = 1.5 \text{ }\mu\text{m}$.

3 Machine-checkable claims and results

C1 — 2θ optic-axis rotation. The two switched states $\phi = \pm\pi/2$ (UP/DOWN) give projected director orientations separated by exactly $2\theta = 45.0^\circ$ (abs. error $7 \times 10^{-15} \text{ deg}$). This is the Clark–Lagerwall rigid-cone claim quoted in the paper; it is a pure geometric consequence of the fixed-cone constraint. **PASS.**

ID	Claim	Result	Status
C1	Switch rotates optic axis by 2θ	45.00° vs 45.0°	PASS
C2	$\tau = \gamma/(P_s E)$ switching-time law	$\tau \propto 1/E$, $R^2=0.9999$	PASS (scaling)
C3	Helix unwinding set by surface/twist ratio, $\sim$ indep. of elastic stiffness (Dahl)	criterion invariant under K -scaling	PASS
C4	Bistability = double-well (elastic+anchor)	2 minima at $\pm 90^\circ$	PASS
C5	Static-friction bistability (Dahl, novel)	memory + threshold switch	PASS

C2 — switching-time law. Overdamped uniform rotation gives a 10–90% switch time linear in $1/E$ ($R^2 = 0.9999$). The fitted slope is $3.5\times$ the naive γ/P_s . This is *not* a failure: the exact analytic 10–90% time equals $\gamma/(P_s E) \cdot \int \sec \phi d\phi$, a field-independent $O(1)$ integration constant (≈ 3.58). Numerical and analytic times agree to $< 3\%$ over the range. The quoted $\tau \propto 1/E$ *scaling* is exact; only the definitional prefactor differs. **PASS (scaling law).**

C3 — helix unwinding, Dahl’s alternative view. Dahl argues (p. 33–34) that surface unwinding of the helix “will occur essentially independently of the strength of the elastic forces, as long as they are weaker than the surface forces.” We hold the dimensionless ratio $W_s/(Kq_0)$ fixed and scale K over two decades: the unwound/wound criterion (sign of ΔF) is **invariant**, and the reduced energy gap collapses to a single value (-11.16). This confirms the transition is governed by the surface-vs-twist competition, not the absolute stiffness. **PASS.**

C4 — double-well bistability. At $E = 0$ the uniform-director energy has exactly two degenerate minima ($\phi = \pm 90^\circ$) with a barrier $2 \times 10^{-4} \text{ J/m}^2$; removing anchoring ($W_s=0$) with $\Delta\epsilon > 0$ destroys the two-state structure. Confirms bistability requires the surface-broken degeneracy the paper discusses. **PASS.**

C5 — static-friction bistability (Dahl’s genuinely novel claim). Dahl proposes (p. 35) that bistability may be held by *static friction* – a threshold/pinning force – “as an alternative or complement to an elastic barrier,” a claim Lagerwall reportedly found “ridiculous.” A Coulomb-friction dynamical model with a *flat* elastic potential retains two stable rest states (memory at $E = 0$), stays pinned below the threshold field ($E_{\text{th}} = 4 \text{ MV/m}$), and switches above it. Dahl’s mechanism is thus self-consistent as a distinct bistability source. **PASS.**

4 Verdict

Coverage: 6/10. The paper is a non-quantitative polemic; there is no headline number, dataset, or figure to reproduce. We replicate the domain physics it argues about (5 distinct claims, including Dahl’s two original alternative-view proposals) but cannot match anything the paper itself computes, because it computes nothing.

Agreement: 8/10. Every checkable statement reproduces: the geometric 2θ result is exact; the $\tau \propto 1/E$ scaling is exact (prefactor explained analytically); the unwinding-invariance, double-well, and static-friction mechanisms all confirm as physically self-consistent. Points withheld because C2’s absolute prefactor is definitional (not stated in the paper) and the model is qualitative, matching a qualitative source.

Overall verdict: REPRODUCED (domain-physics level). The source is non-replicable as a primary paper (correctly flagged), but its physical content is sound and every reduced claim checks out.